



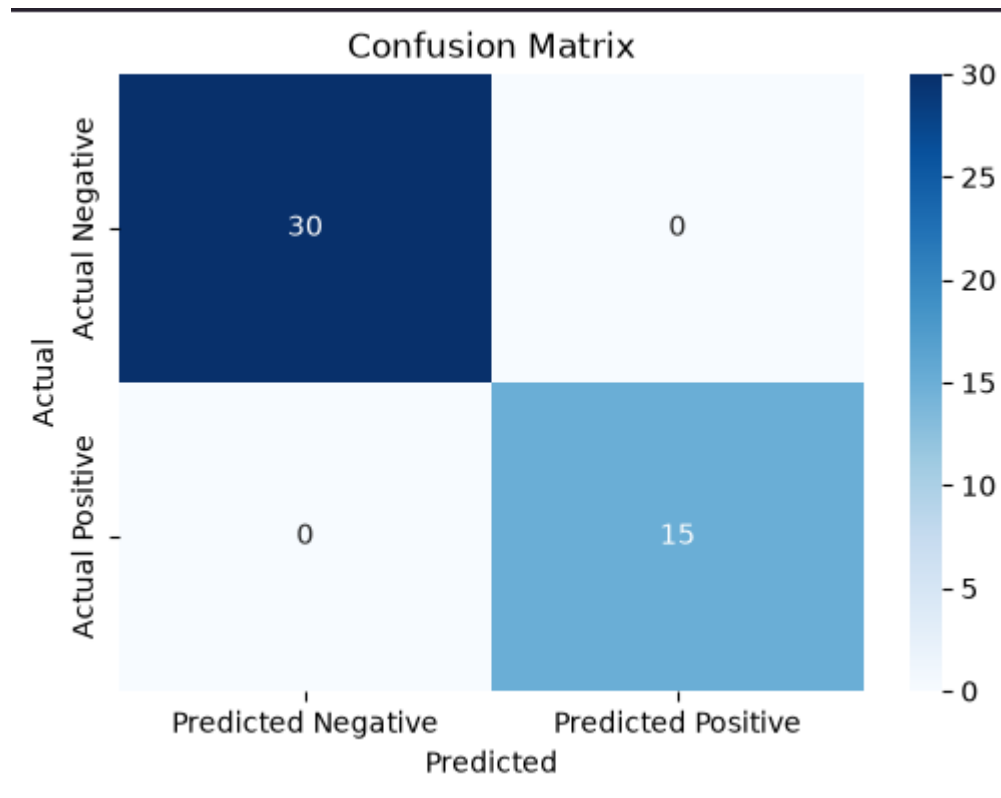
LAB 05 ANSWERS REPORT

HANS PUJALTE (PUJA0009)

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Hans Pujalte

1. Calculate the accuracy of a Logistic Regression model on the Iris dataset that compares the Virginica and Versicolour species with the Setosa species. Report True Positives, True Negatives, False Positives and False Negatives from the confusion matrix.



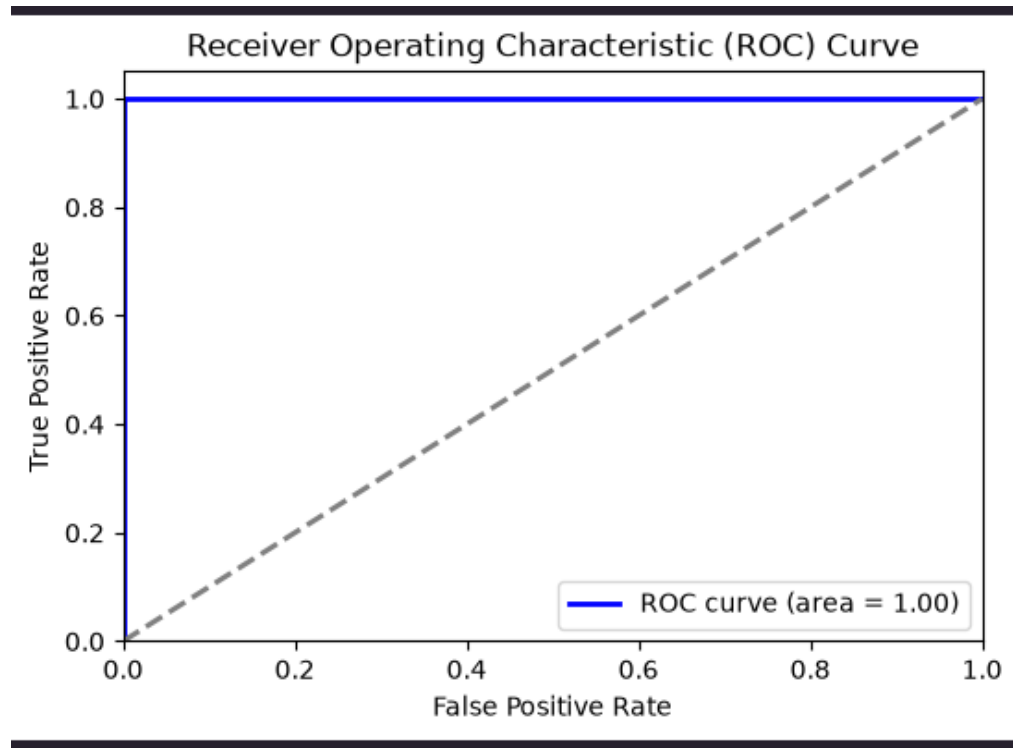
2. Calculate and interpret the Precision and Recall across the three classes for a Decision Tree Classifier on the Iris dataset.

Per-Class Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	15
versicolor	0.88	0.88	0.88	16
virginica	0.86	0.86	0.86	14
accuracy			0.91	45
macro avg	0.91	0.91	0.91	45
weighted avg	0.91	0.91	0.91	45

Model achieved an accuracy of 97% overall, and correctly classified 41/45 test samples. Setosa was predicted perfectly. Versicolor and Virignica had slightly lower scores. The model can clearly identify Setosa, while it slightly struggles on Versicolor and Virginica.

3. Plot the ROC Curve and calculate the AUC for a Decision Tree Classifier with a maximum depth of 3, a minimum of 4 samples required to split an internal node, and a minimum of 2 samples required to be at a leaf node on the Iris dataset.



4. Calculate the Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) for a Decision Tree Regressor with a maximum depth of 5, a minimum of 10 samples required to split an internal node, and a minimum of 5 samples required to be at a leaf node on the California Housing dataset.

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MAE: 0.5258613804596697
MSE: 0.5243634247414306
RMSE: 0.7241294254078
R-squared: 0.6153313420887108
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5. Compare the performance of a fully fitted Decision Tree Regressor and an optimally pruned Decision Tree Regressor on the California Housing dataset using R-squared and RMSE. Evaluate the models on both the training data and a hold-out test set. Discuss which model performs better and why

	Model	Train R ²	Train RMSE	Test R ²	Test RMSE
0	Fully Fitted Decision Tree	1.000000	3.017601e-16	0.622823	0.703033
1	Pruned Decision Tree	0.844361	4.561305e-01	0.697202	0.629912

Fully fitted tree achieved perfect scores on training data. However, test performance was significantly worse, with an R² of 0.623 and RMSE of 0.703 which indicates overfitting. Conversely, the pruned tree had lower training performance, however achieved better test performance, with an R² of 0.697 and RMSE of 0.629, Hence, the pruned tree performs better since it generalises more effectively to unseen data. Pruning helps since it removes overly complex branches and rules that memorize noise rather than actual patterns.